AI Math Tutor

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Project Description

Solving algebra problems from textbooks or handwritten notes can be challenging for students, especially when they struggle with symbol recognition, equation interpretation, or step-by-step reasoning. Manual solving often leads to common mistakes such as sign errors, incorrect transposition, and arithmetic miscalculations.

AI Math Tutor addresses these challenges by providing an intelligent, image-based algebra problem-solving platform. The system allows students to upload a photo of an algebra equation, which is automatically processed using Optical Character Recognition (OCR) and Vision-Language Models to extract mathematical symbols and convert them into LaTeX format. The extracted equation is then solved using symbolic mathematics, and a detailed step-by-step solution is generated.

In addition, the platform identifies common mistake patterns and highlights potential trap areas, helping students understand not just the final answer but also the reasoning process. AI Math Tutor transforms traditional problem-solving into an interactive and personalized learning experience.

Use Case Scenarios

Scenario 1: Image-Based Algebra Problem Recognition

Students upload a photo of a handwritten or printed algebra problem. The system analyzes the image, extracts mathematical symbols, and converts the problem into a structured LaTeX equation.

Output:

Recognized equation

Clean LaTeX preview

Scenario 2: Equation Solving and Computation

The extracted equation is parsed and solved using symbolic computation techniques. The system supports linear and quadratic equations and performs accurate numerical evaluation.

Output:

Simplified equation

Computed solution

Scenario 3: Step-by-Step Solution Generation

The platform generates a logical, easy-to-understand step-by-step solution, breaking down the problem into intermediate steps.

Output:

Step-wise explanation

Final result

Scenario 4: Mistake Identification and Feedback

The system analyses the solving process and checks for common algebraic mistakes such as sign errors, incorrect distribution, and arithmetic traps.

Output:

Mistake alerts

Explanation of error type

Architecture Overview

AI Math Tutor is an intelligent, modular educational platform designed to solve algebra problems from images using artificial intelligence and symbolic computation. The system is built using the Streamlit framework for rapid user interaction and integrates OCR, Vision-Language Models, and symbolic mathematics engines to deliver accurate, step-by-step solutions.

The platform processes image-based inputs through multiple layers, including image preprocessing, symbol recognition, equation interpretation, mistake detection, and explanation generation. By combining computer vision with mathematical reasoning, AI Math Tutor provides both correct solutions and conceptual clarity.

Core Technologies

Streamlit: Web framework for building interactive user interfaces and handling image uploads.

Vision-Language Model (Gemini / OpenAI Vision): Converts extracted mathematical symbols into structured LaTeX equations.

OCR Engine (EasyOCR / Tesseract): Extracts handwritten or printed text from uploaded images.

Python: Core backend programming language.

SymPy: Symbolic mathematics library for solving algebraic equations.

OpenCV: Image preprocessing and enhancement.

MathJax / LaTeX Renderer: Displays equations and solutions in a readable mathematical format.

Component-Wise Architecture

Component	Description
Frontend UI (Streamlit)	Interactive web interface that allows users to upload algebra problem images and view solutions. Displays LaTeX equations, step-by-step solutions, and mistake alerts.
Image Processing Module	Preprocesses images using OpenCV by resizing, noise reduction, and contrast enhancement to improve OCR accuracy.
OCR Layer	Extracts mathematical symbols and expressions from images using EasyOCR or Tesseract.
Vision AI Layer	Converts OCR output into structured LaTeX equations. Understands mathematical context and formatting.
Equation Solver Module	Uses SymPy to parse LaTeX equations and compute symbolic and numeric solutions.
Mistake Detection Module	Identifies common algebraic errors such as sign mistakes, incorrect distribution, and arithmetic traps.
Solution Generator Module	Produces a clear step-by-step explanation of how the equation is solved.
Result Display Engine	Renders equations, steps, final answers, and mistake feedback using MathJax within Streamlit.

Pre-requisites

Software Requirements

1. Python 3.9 or above
2. Streamlit Framework
3. OpenCV
4. EasyOCR / Tesseract OCR
5. SymPy
6. Pillow, NumPy
7. Gemini API / OpenAI Vision API Key

Hardware Requirements

Laptop or Desktop System
Camera or Mobile Phone (for capturing algebra problem images)
Stable Internet Connection

Project Workflow

1. Gemini AI Setup & Initialization

Activity 1.1: Set up Google AI Studio access with Gemini 2.5 Flash API

Create or sign in to Google AI Studio account
Generate API key for Gemini Vision access
Enable image + text input capabilities
Test API connectivity with sample algebra images

Activity 1.2: Configure API keys and environment

Create .env file for secure API key storage
Set GEMINI_API_KEY as environment variable
Configure request limits and timeout handling
Validate API responses for image-to-LaTeX conversion

Activity 1.3: Fine-tune prompts for mathematical understanding

Design prompts for converting image text to LaTeX
Create prompts for step-by-step algebra solving
Develop prompts for identifying common algebra mistakes
Test prompt accuracy with handwritten and printed equations

2. Core Functionality Development

Activity 2.1: Design Streamlit application structure

Create modular file structure for UI and backend logic
Implement Streamlit page layout and navigation
Set up session state management
Enable image upload and preview

Activity 2.2: Implement equation processing pipeline

Image preprocessing using OpenCV
OCR extraction using EasyOCR / Tesseract
Vision AI conversion of OCR output to LaTeX
Validate extracted mathematical expressions

Activity 2.3: Integrate Gemini Vision prompts

Send image and OCR output to Gemini Vision model
Parse structured LaTeX responses
Handle malformed or incomplete outputs
Ensure consistent equation formatting

3. Backend Implementation (app.py)

Activity 3.1: Implement equation solver logic

Parse LaTeX equations into SymPy expressions
Solve linear and quadratic equations
Simplify symbolic expressions
Handle invalid or unsolvable equations

Activity 3.2: Develop mistake detection module

Detect sign errors
Detect incorrect distribution
Detect arithmetic calculation mistakes
Flag common student trap areas

Activity 3.3: Generate step-by-step solutions

Break down solution into logical steps
Generate explanations for each step
Highlight correction points if mistakes are detected
Format steps for UI display

Activity 3.4: Add utility functions

Image validation and resizing
Error handling and fallback responses
Logging for debugging and monitoring
Confidence scoring for OCR and solution accuracy

4. UI Development

Activity 4.1: Build interactive Streamlit interface

Home page with project overview
Image upload section
Equation preview (LaTeX rendering)
Solve button and progress indicators

Activity 4.2: Implement solution display components

Step-by-step solution cards
Final answer highlighting
Mistake warning alerts
Mathematical formatting using MathJax

Activity 4.3: Enhance user experience

Loading spinners during AI processing
Error messages for invalid inputs
Clear instructions for users
Mobile-responsive layout

5. Testing & Optimization

Activity 5.1: Perform comprehensive testing

Test with handwritten equations
Test with printed textbook equations
Validate OCR accuracy
Validate solver correctness

Activity 5.2: Evaluate AI response quality

Accuracy of LaTeX conversion
Correctness of mathematical solutions
Clarity of step-by-step explanations
Effectiveness of mistake detection

Activity 5.3: Optimize prompts and performance

Improve prompt clarity and structure
Reduce unnecessary API calls
Handle edge cases (blurred images, partial equations)
Implement retry logic for failed AI responses

Activity 5.4: Ensure solution quality and reliability

Validate consistency of results
Improve explanation readability
Ensure UI stability
Prepare application for deployment

MILESTONE 1: Gemini AI Initialization

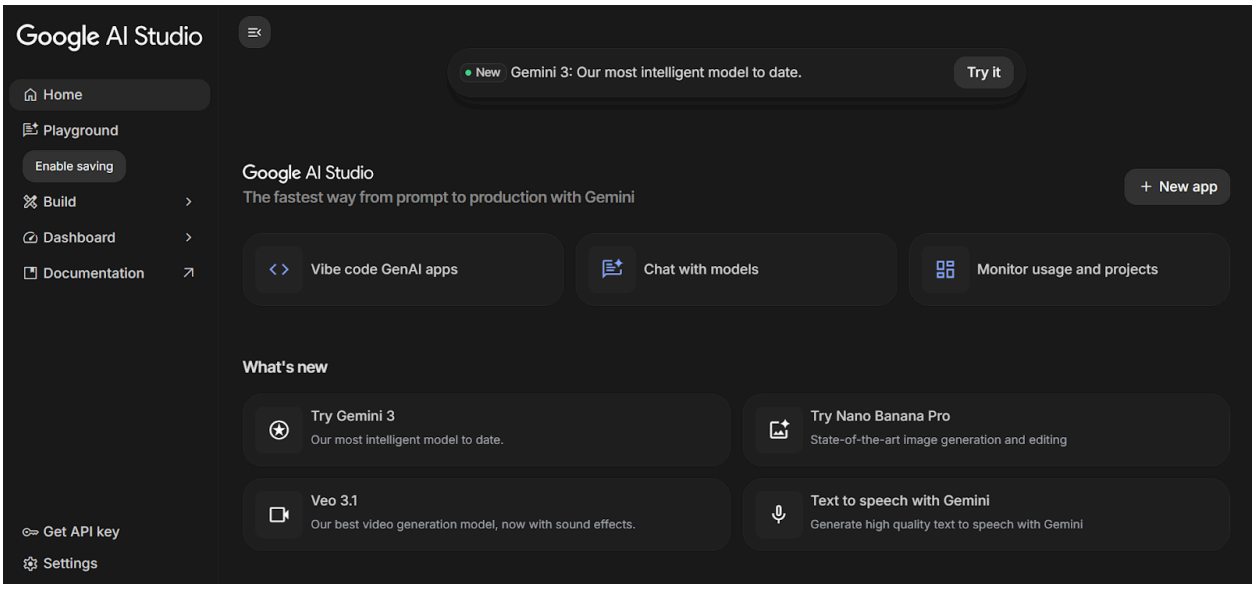
In this foundational stage, the objective is to establish and configure the core AI infrastructure required to power the **AI Math Tutor** application. The platform relies on **Gemini 2.5 Flash**, Google's fast and efficient multimodal foundation model, to interpret algebra problem images, convert them into LaTeX equations, and assist in step-by-step mathematical reasoning.

This milestone ensures reliable AI connectivity, secure API management, and validated response quality before core feature development begins.

Activity 1.1: Set up Google AI Studio and enable Gemini API

1. **Sign in to Google AI Studio**
   Visit: https://makersuite.google.com/app/apikey
   Sign in using a Google account
   Accept the terms and conditions

1. **Create API Key**
   Click on **"Create API Key"**
   Copy the generated API key
   Store the API key securely for later use



3. **Select Model**

   Choose **Gemini 2.5 Flash** from the available models

   Review model specifications:

   - Supports multimodal inputs (image + text)
   - Fast response time
   - Suitable for educational and reasoning-based applications
   - Efficient for step-by-step math explanations

Activity 1.2: Configure environment and test connection

1. **Create `.env` file**

Create a .env file in the project root directory to securely store API credentials.

```
GEMINI_API_KEY=your_api_key_here
SECRET_KEY=your_secret_key_here
```

2. **Install dependencies**

Install the Gemini SDK and environment management libraries.

```
pip install google-generativeai python-dotenv
```

3. **Test API Connection**

Verify that the Gemini API is correctly configured and accessible.

```
import google.generativeai as genai
import os
from dotenv import load_dotenv

load_dotenv()

genai.configure(api_key=os.getenv("GEMINI_API_KEY"))

# List available models
for model in genai.list_models():
    if "generateContent" in model.supported_generation_methods:
        print(f"- {model.name}")
```

Expected Output:

A list of Gemini models, including models/gemini-2.5-flash.

Activity 1.3: Validate connectivity and response quality

Test Text Generation (Math-Oriented)

Test Gemini's ability to understand and respond to algebra-related prompts.

```
model = genai.GenerativeModel("models/gemini-2.5-flash")

response = model.generate_content(
    "Solve the equation 2x + 5 = 15 and explain the steps."
)

print(response.text)
```

MILESTONE 2: Core Functionalities Development

This milestone focuses on building the core backend intelligence using **Flask** and **Gemini 2.5 Flash**. The primary goal is to develop a robust system capable of **accepting algebra problem images**, **converting them into LaTeX equations**, **solving equations**, and **detecting common mistakes**, while returning **clear step-by-step solutions** suitable for students.

File Explorer Structure

```
ai_math_tutor/
│
├── app.py                  # Main Flask application
├── requirements.txt        # Python dependencies
├── .env                    # Environment variables
├── .env.template           # Environment variable template
│
├── static/
│   ├── css/
│   │   └── style.css       # UI styling
│
├── templates/
│   ├── index.html          # Upload page
│   ├── solution.html       # Step-by-step solution card
│
├── utils/
│   ├── ocr_utils.py        # Image to LaTeX conversion
│   ├── solver.py           # Equation solving logic
│   ├── mistake_check.py    # Error and trap detection
│
├── outputs/
│   ├── results/            # Generated solutions
│
└── tests/
    └── test_api.py         # API test cases
```

Activity 2.1: Flask Initialization and Import Libraries

Import Required Libraries

```python
from flask import Flask, request, jsonify, render_template
import google.generativeai as genai
import os
from dotenv import load_dotenv
from datetime import datetime
```

Flask Initialization

```python
load_dotenv()

app = Flask(__name__)
app.config['SECRET_KEY'] = os.getenv("SECRET_KEY")

# Configure Gemini API
genai.configure(api_key=os.getenv("GEMINI_API_KEY"))
model = genai.GenerativeModel("models/gemini-2.5-flash")
```

Activity 2.2: Main Functionality Code Routing

```python
@app.route('/api/solve-equation', methods=['POST'])
def solve_equation():
    image = request.files.get('image')

    prompt = """
    Convert the algebra problem in the image into LaTeX,
    solve it step-by-step, and identify common mistakes
    students might make.
    """

    response = model.generate_content([prompt, image])
    content = response.text

    return jsonify({
        "success": True,
        "solution": content
    })
```

Question Bank Generation

```python
@app.route('/api/generate-question-bank', methods=['POST'])
def generate_question_bank():
    data = request.json
    subject = data.get('subject')
    chapters = data.get('chapters', [])
    num_questions = data.get('num_questions', 30)

    prompt = f"""Create {num_questions} original questions
    for {subject} covering: {', '.join(chapters)}"""

    response = model.generate_content(prompt)
    content = clean_json_response(response.text)

    return jsonify({'success': True, 'content': content})
```

Assignment Generation

```python
@app.route('/api/generate-assignment', methods=['POST'])
def generate_assignment():
    data = request.json
    subject = data.get('subject')
    topic = data.get('topic')
    grade = data.get('grade')
    assignment_type = data.get('type', 'Project')

    prompt = f"""Create a {assignment_type} for Grade {grade}
    in {subject} on "{topic}" with objectives, instructions,
    materials, rubric, and real-world connection."""

    response = model.generate_content(prompt)
    content = clean_json_response(response.text)

    return jsonify({'success': True, 'content': content})
```

Activity 2.3: Export Functionality Routing

Symbol Mapping (Image → LaTeX)

```python
@app.route('/api/image-to-latex', methods=['POST'])
def image_to_latex():
    image = request.files.get('image')

    prompt = "Convert the handwritten or printed algebra problem into LaTeX format."
    response = model.generate_content([prompt, image])

    return jsonify({
        "success": True,
        "latex": response.text
    })
```

Equation Solver

```python
@app.route('/api/compute-solution', methods=['POST'])
def compute_solution():
    data = request.json
    equation = data.get("equation")

    prompt = f"Solve the equation {equation} step-by-step with explanations."
    response = model.generate_content(prompt)

    return jsonify({
        "success": True,
        "steps": response.text
    })
```

Mistake Detection Module

```python
@app.route('/api/mistake-check', methods=['POST'])
def mistake_check():
    data = request.json
    equation = data.get("equation")

    prompt = f"""
    Analyze the equation {equation} and identify
    common student mistakes or trap areas.
    "  Ask ChatGPT

    response = model.generate_content(prompt)

    return jsonify({
        "success": True,
        "mistakes": response.text
    })
```

Activity 2.3: Export Functionality Routing

PDF Export

```python
@app.route('/api/export-pdf', methods=['POST'])
def export_pdf():
    data = request.json
    content = data.get("solution")

    filename = f"outputs/solution_{datetime.now().strftime('%Y%m%d_%H%M%S')}.pdf"
    # PDF generation logic

    return jsonify({
        "success": True,
        "file": filename
    })
```

Word Export

```python
@app.route('/api/export-word', methods=['POST'])
def export_word():
    data = request.json
    content = data.get("solution")

    filename = f"outputs/solution_{datetime.now().strftime('%Y%m%d_%H%M%S')}.docx"
    # Word generation logic

    return jsonify({
        "success": True,
        "file": filename
    })
```

Activity 2.4: Utility Functions

JSON Response Cleaner

```python
def clean_response(text):
    text = text.replace("```", "")
    return text.strip()
```

Filename Generator

```python
def generate_filename(prefix, extension):
    timestamp = datetime.now().strftime("%Y%m%d_%H%M%S")
    return f"outputs/{prefix}_{timestamp}.{extension}"
```

MILESTONE 3: Backend – Flask Integration (app.py)This

milestone focuses on implementing the **complete Flask backend** for the **AI Math Tutor** application. It integrates all previously developed AI modules, ensures secure API routing, handles errors gracefully, and supports export-ready outputs. The backend enables seamless communication between the user interface and the AI-powered math processing engine.

Activity 3.1: Complete Route Definitions

All backend routes are fully defined and validated to ensure reliable system behavior. Each API endpoint includes:

Input validation for image and equation data
Proper error handling and fallback responses
Logging for debugging and monitoring
Structured JSON response formatting

Defined API Routes

/api/image-to-latex – Converts uploaded algebra problem images into LaTeX format
/api/solve-equation – Generates step-by-step algebra solutions
/api/mistake-check – Identifies common student mistakes and trap areas
/api/export-pdf – Exports solutions as formatted PDF files
/api/export-word – Exports solutions as Word documents

Activity 3.2: Modular Architecture

Main Application Structure

The Flask backend follows a **modular architecture** to improve maintainability, scalability, and debugging efficiency. Core logic is separated into independent utility modules.

```python
if __name__ == "__main__":
    print("=" * 60)
    print("AI Math Tutor - Backend Server Starting")
    print("=" * 60)
    print("✓ Gemini API Key configured")
    print("✓ Using model: gemini-2.5-flash (FREE TIER)")
    print("\nServer running at http://localhost:5000")
    print("=" * 60)

    app.run(debug=True, host="0.0.0.0", port=5000)
```

Backend Module Responsibilities

ocr_utils.py – Image preprocessing and symbol recognition
solver.py – Step-by-step algebra solving logic
mistake_check.py – Detection of common algebra mistakes
export_utils.py – PDF and Word file generation
response_utils.py – JSON cleanup and formatting

This structure ensures **clean separation of concerns** and enables future expansion to support calculus or geometry.

Activity 3.3: Export Implementation

The AI Math Tutor supports professional-quality output formats for academic use.

Export Features Implemented
PDF generation with mathematical formatting and solution steps
Word document creation with headings and readable equations
Timestamp-based file naming for uniqueness
Automatic file cleanup and storage management

Supported Export Formats

📄 PDF (for assignments and print use)
📝 Word (.docx) (for editing and submission)

Error Handling & Response Management

The backend includes robust error-handling mechanisms:

Invalid image uploads detection
Empty or unclear equation handling

Gemini API failure fallback messages
Standardized success and error JSON responses

This ensures a **stable learning experience for students** even under edge cases.

MILESTONE 4: UI Development

This milestone focuses on designing and implementing an intuitive, responsive, and student-friendly frontend for the **AI Math Tutor** application using **Streamlit, HTML, CSS, and basic JavaScript concepts (where applicable)**. The UI enables users to upload algebra problem images, view step-by-step solutions, and understand mistake-prone areas clearly.**Activity 4.1: Home Page Development**

Landing Page Features:

Hero section with project description
Feature cards for each generator
Easy navigation
Responsive design

Activity 4.1: Home Page Development

Landing Page Features

Hero section with **AI Math Tutor** project overview
Clear description of image-based algebra solving
Feature cards highlighting core functionalities
Simple and intuitive navigation
Fully responsive design for desktop and mobile users

Home Page Feature Highlights

Upload Algebra Image – Allows students to upload handwritten or printed algebra problems
LaTeX Conversion – Displays converted mathematical expressions
Step-by-Step Solution – Shows structured reasoning
Mistake Detection – Highlights common algebra traps

Activity 4.2: Core Functional Pages

1. Algebra Problem Solver Page

UI Components:

Image upload field (PNG/JPG)
Preview of uploaded image
Converted LaTeX equation display
"Solve Problem" button
Loading indicator during AI processing

Displayed Output:

Recognized equation in LaTeX format
Step-by-step solution cards
Final numeric answer

2. Step-by-Step Solution Display

Features:

Clearly numbered steps
Mathematical formatting using LaTeX rendering
Easy-to-read explanation text
Section-wise separation of steps

3. Mistake Detection & Hint Section

UI Elements:

Highlighted mistake-prone steps
Common algebra traps (sign errors, incorrect transposition, wrong factorization)
Friendly hints and corrective guidance

Activity 4.3: Result Presentation and Interaction

Formatted Output Display

Card-based layout for solutions
Expand/collapse sections for steps
Clean typography for mathematical symbols
Visual emphasis on final answer

User Interaction Enhancements

Real-time loading spinner
Error messages for invalid images
Success notifications on completion
Smooth page refresh after submission

Activity 4.4: UI Styling and Responsiveness

Design Enhancements

Consistent color theme for educational clarity
High contrast for mathematical readability
Clean spacing and alignment
Mobile-friendly responsive layout

Accessibility Considerations

Large readable fonts
Clear button labels
Simple language for explanations
Minimal visual clutter

MILESTONE 5: Testing & Optimization

This milestone focuses on validating the correctness, reliability, and performance of the **AI Math Tutor** system. Comprehensive testing is performed to ensure accurate algebra recognition, step-by-step problem solving, and effective mistake detection across different difficulty levels.

Activity 5.1: Comprehensive Testing

Test Scenarios

1. Algebra Problem Solver Testing

Subject: Mathematics
Topic: Algebra (Linear Equations)
Difficulty: Medium
Input Type: Image-based problems

Expected Output:

Correct LaTeX equation extraction
Accurate step-by-step solution
Correct final numeric answer

Test Cases Include:

Simple linear equations
Multi-step algebraic equations
Equations with fractions and negative values

Handwritten and printed algebra images

3. Edge Case Testing

Blurry or low-quality images
Incomplete equations
Non-algebra images
Unsupported symbols

Expected Behavior:

Graceful error handling
User-friendly feedback messages
No system crashes

Activity 5.2: Quality Evaluation

Evaluation Criteria

Solution Accuracy: Correctness of final answer
Step Clarity: Logical and easy-to-follow explanation
LaTeX Accuracy: Proper mathematical formatting
Mistake Identification: Relevance and clarity of detected errors
User Readability: Student-friendly language

Result Analysis

High accuracy for clean algebra images
Clear step-by-step reasoning suitable for learners
Effective identification of common algebra mistakes
Consistent output formatting across different inputs

Activity 5.3: Performance Optimization

Improvements Implemented

Optimized prompt design for faster reasoning
Reduced unnecessary API calls
Efficient image preprocessing before inference
Improved loading indicators during processing

System Performance Enhancements

Faster response time for equation solving
Reduced latency in image-to-LaTeX conversion

Stable performance under multiple user requests

Activity 5.4: Reliability and Usability Validation

Usability Testing

Easy image upload process
Clear navigation and feedback
Smooth interaction for students

Reliability Checks

Consistent results for repeated inputs
Proper exception handling
Stable application behavior

Conclusion

The **AI Math Tutor** project shows how Generative AI can be used in education to help students solve algebra problems easily. It provides an intelligent and interactive learning experience for students. The system allows users to upload images of handwritten algebra problems. The application then converts these images into proper mathematical expressions using symbol mapping. After conversion, the system solves the equations step by step. It also explains the solution clearly so students can understand the process.

In addition, the system identifies common mistakes made by students and provides corrective guidance. This helps learners not only find the correct answer but also improve their understanding of algebra concepts. The project uses a modular and scalable architecture built with the **Streamlit framework** and **Gemini 2.5 Flash**. This ensures fast responses and accurate problem solving.

Testing and optimization were performed to ensure the system is reliable and performs well. The project demonstrates how AI can support modern digital learning. It helps students study at their own pace and improve their problem-solving skills.

Overall, the AI Math Tutor acts as a powerful educational assistant. It also provides a strong base for future improvements such as multi-subject support, adaptive learning, and personalized student analysis.